Marmara Üniversitesi
Electrical and Electronics Engineering Department
Spring 2024 – EE 4051 – Quiz 2
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Name: KEY Quiz (2) ID: EE 4051 Mark /30

1) A lossy transmission line has the following parameters: R'=0.404 (mΩ/m), L'=2.0 (μH/m), G'=0 and C'=5.56 (pF/m). For operation at 5 kHz, determine (a) the line propagation constant (b) the attenuation constant, (c) the phase constant, (d) the phase velocity and (e) the characteristic impedance.

a) $\gamma^2 = (R' + j\omega L')(0 + j\omega C')$

$\Rightarrow \quad \gamma = 3.37 \times 10^{-7} + j\,1.05 \times 10^{-4}$ (1/m)

b) $\alpha = 3.37 \times 10^{-7}$ Np/m

c) $\beta = 1.05 \times 10^{-4}$ rad/m

d) $v = \omega/\beta = 2\pi \times 5 \times 10^3 / 1.05 \times 10^{-4} = 3 \times 10^8$ m/s

e) $Z_0 = \dfrac{R' + j\omega L'}{\gamma} = 600 - j2$ (Ω)

2) Consider a 50-Ohm coaxial cable that is 4.2 m long and operating at a frequency of 300 MHz. The line is terminated by a load of 80−*j*60 (Ohms). Find (a) the input impedance at the generator side, (b) the voltage reflection coefficient at the load and (c) the voltage standing wave ratio. Take v=0.8c.

a) $\beta = \omega/v = 2\pi \times 300 \times 10^6 / 0.8 \times 3 \times 10^8 = 7.85$ rad/m

50-ohm line $\Rightarrow Z_0 = 50\ \Omega$: Lossless

$$Z_{in} = Z_0 \left(\frac{Z_L + j Z_0 \tan(\beta l)}{Z_0 + j Z_L \tan(\beta l)} \right) = 24.4 \angle 35.7^\circ \ (\Omega)$$

b) $\Gamma = \dfrac{Z_L - Z_0}{Z_L + Z_0} = \dfrac{80 - j60 - 50}{80 - j60 + 50} = 0.47 \angle -36^\circ$

c) $s = \dfrac{1 + |\Gamma|}{1 - |\Gamma|} = \dfrac{1 + 0.47}{1 - 0.47} = 2.77$